6COSC023W

Computer Science Final Project

Final Year Project (FYP) - Report

Project Title

Student: first and last name (student number)

Supervisor: first and last name

Degree: BSc (Hons) XXX

This report is submitted in partial fulfillment of the requirements for the

BSc (Hons) Computer Science degree

BEng Software Engineering degree

School of Computer Science & Engineering

University of Westminster

Date

Document Scope

The purpose of this document is to describe and reflect on the processes that took place in developing the Final Project. Discuss any ethical issues associated with your project and describe the methodology that was adopted to develop its design, implementation and testing.

All chapter word counts in this document are approximate and are not intended to be prescriptive.

All sections in orange (like this one) must be deleted and removed before submitting the report.

Declaration

This report has been prepared based on my own work. Where other published and unpublished source materials have been used, these have been acknowledged in references.

Word Count: The final word count
Student Name: Your full name
Date of Submission: Submission date

This is an important section!
Add the updated word count (do not count words in the Acknowledgments, Table of Contents, Table of Figures, Table of Tables, References, Bibliography and Appendix). Add your name and the date of submission.

Abstract

500 words

Summarise here the problem statement and the project aim(s). Provide a brief description of the methodology followed, technologies/algorithms used, the main solution/results, your conclusions and future recommendations.

Acknowledgements

Thank those who helped you build your project and supported you during its development, testing and evaluation.

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List of Figures

Provide a list of figures, linking figure numbers to page numbers. If you can, hyperlink the page numbers/figures. If you have no figures in your document, please remove this page.

Figure 1. Add a caption explaining the image here. .....**Error! Bookmark not defined.**

List of Tables

Provide a list of tables (if any), linking table numbers to page numbers. If you can, hyperlink the page numbers/tables. If you have no tables in your document, please remove this page.

Table 1. Add caption here.....**Error! Bookmark not defined.**

1. Introduction

Introduce the project, significance, include the problem statement, project aim(s) and objectives.

1.1 Problem statement

500 words

Give some background on the problem you intend to solve and justify the need for the software/application. Use references to support your statements, when possible, statistics, illustrations, diagrams, and figures.

1.2 Aims and Objectives

300 words

The aim(s) describe, in a few sentences, the overarching purpose(s)/intention(s) of the software/application. What is the point of developing the software/application, what you wish to achieve?

Objectives describe with some detail the individual steps you will take to fulfil the project aim(s).

2. Background

Include a literature survey on the topic, discuss existing similar or relevant applications to yours and the result of a critical review of tools, techniques, theories, and algorithms that are used to tackle projects like yours.

2.1 Literature survey

800 words

Describe the initial results of a literature survey on a selected research topic or application area related to your project subject. Use relevant books, published research articles, as well as credible Internet content for the purpose. Make use of in-text references to indicate your sources.

2.2 Review of projects/applications

800 words

Describe your background research on existing projects/software/applications, tools/frameworks/methods/algorithms/techniques relevant to your project and their advantages and disadvantages. Use illustrations, diagrams, and screenshots for the purpose.

You may present the Gap Analysis Table (Table of Features in this section, comparing the main features of the above projects/software/applications and the one you developed).

Features	Existing Software 1	Existing Software 2	Existing Software 3	Existing Software 4	Existing Software 5	My Solution
Feature 1	YES	NO	YES	NO	NO	YES
Feature 2	NO	YES	NO	YES	YES	YES
Feature 3	NO	NO	NO	NO	YES	YES
Feature 4	NO	YES	NO	NO	NO	YES
Feature 5	NO	NO	NO	NO	NO	YES

A comparison table may also be provided to distinguish the key characteristics of features/methods/algorithms/techniques relevant to your project.

2.3 Review of tools, frameworks and techniques

800 words

Describe the results of a survey on relevant tools/frameworks that can be used to develop applications, such as the one you built for your project, including algorithms, programming languages, environments and libraries. List their advantages and disadvantages to justify your selection. Use illustrations, diagrams, and screenshots to support your purpose.

3. Legal, social, sustainability and ethical issues

500 words

Consider any legal, ethical, social, professional, sustainability and security issues associated with your research and the software/application you are building and/or the data you are collecting/analysing.

4. Methodology

1000 Words

Describe the life cycle stages of the project, methodology, and development techniques you followed in the design and implementation of your project.

As examples: Gantt chart for life cycle, Waterfall or Agile for development methodology. Use an appropriate methodology for the project and list the key steps and milestones.

Discuss the implementation of your project and your considerations for UX and UI. Describe your testing methodology and give adequate examples, e.g., unit testing for typical client-server applications, white box for algorithmic and mission-critical code, etc. Discuss why your chosen methodology is suitable for the project.

Please note that even if you are using an Agile methodology, you will still need to provide a high-level waterfall plan with key milestones, along with any Agile iterations, detailed in this report.

5. Design

Describe your final software structure using diagrams where necessary.

1200 Words

Discuss in some detail (if relevant) issues relating to:

- *High-level architecture*
- *Infrastructure*
- *Conceptual diagrams*
- *User Interfaces*
- *Functionality*
- *Algorithm development*
- *Content creation*
- *Other*

Discuss how those address the project requirements.

Use appropriate design methods for your project and extend your design to include implementation details that were not included in your Interim Progress Demonstration (IPD). e.g., make use of UML, such as class diagrams, sequence/activity/state diagrams, for complex algorithms and workflows. Use UI design methodology and heuristics for predominantly UX-based projects. If you intend to develop an app/software/dashboard, you may need to use/create ERD, flowcharts, storyboards, and prototypes. It is up to you to use the appropriate design that best describes your implementation.

6. Tools and implementation

6.1 Tools

500 words

Describe the tools (programming environments & languages, frameworks, and libraries) you used for the development of your application. Justify your choices with references to your use cases or list of requirements.

State existing skills development and any new skills you employed for building your project.

6.2 Implementation

2500 words

Explain the implementation of the main code by use case. Include pseudocode or snippets of any novel code. Highlight any code that is adopted/adapted and give the original sources. Make references to your design documentation where appropriate.

7. Testing

Create sufficient test cases to ensure that the applications meet the requirements and function correctly.

7.1 Test coverage

800 words

Discuss black box and/or white box testing in relation to the requirements. Include specific test cases labelled by the relevant requirements. Use a standard test plan template to document test results.

7.2 Test methodology

800 words

Describe how the output was tested and why. Discuss how you obtained and used feedback, using expert and/or non-expert users.

8. Conclusions and reflections

1000 words

Provide critical reflections on ALL aspects of the project lifecycle. Include conclusions on the resulting application, research, and findings. Reflect on each aspect of your project life cycle. Critically evaluate how effectively your results meet your stated objectives. Reflect on the strengths and weaknesses of your implementation, discuss the acquisition of any new knowledge and skills and consider further work. Mention future recommendations.

9. References

Include a list of items cited in your text (books, papers, websites, etc.). Use the ***Harvard referencing style****, for this purpose*.

10. Bibliography

Include here a list of general reading items (books, papers, websites, etc.). List the items in alphabetical order, using Harvard style to describe them.

Appendix

Depending on your project, useful information which is not presented in the main report could be attached to the Appendix. For example,

Appendix 1 - Provide a table that outlines refined project requirements, including both functional and non-functional requirements.

Appendix 2 - Provide a link to an online video demo of the project.

Appendix 3 – Provide the screenshots of all the implemented user interfaces / results.

Appendix 4 – Provide the most challenging code segments (Do not include the entire code). You may include a link to your GitHub project repository.

Any additional material, if appropriate, should be included in separate appendices.